Misorientation accumulation along columnar grains in GammaPrint-700 laser powder bed fusion: 3D-EBSD (Lamb et al.) versus SPPARKS

SURGE-kokkos branch, `additive/texture/kk` Moser deck

Revision 2, 2026-09-09

Abstract

Lamb et al. measured the starting reference orientation deviation (SROD) of columnar grains in a 3D-EBSD volume of laser powder bed fusion GammaPrint-700 (Scripta Materialia 2024) and, at higher resolution, within grains and cells (Met. Trans. A 2026). Re-analysing their population curves shows statistics consistent with a random-axis rotation walk, $\mathrm{RMS}^2 = Dh$ with $D \approx 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$ against build height and a 5/95 width of about 1.4 times the mean, the Maxwell value; the published curves do not distinguish this from the authors' own two-regime reading over most of the height range, and the mechanism (Lamb: mushy-zone shrinkage and thermomechanical cycling in the heat-affected zone) is not identified by the statistics. Their example grains rotate along a directed path. SPPARKS `additive/texture` and `additive/texture/kk` carry a phenomenological two-term model applied at capture: a per-voxel walk inherited along each lineage and a coherent per-grain walk of the rotation vector, indexed by inheritance path length and correlated between levels. A Moser-source GP700 deck calibrated on the 2024 originating-grain curve (in-sample RMS 0.34°) reproduces a target it was not tuned on, the established-grain curve of the same paper (two builds: RMS 0.28 and 0.47°, bias −0.11 and 0.25°), which the per-voxel model alone fails (bias 3.0°); against the modal SROD trajectory of the 2026 dataset over its first 120 µm the two models are comparable, the per-voxel model closer below 60 µm. Per grain, the model's accumulation rate (mean 0.84° per 30 µm) is half the 2026 average and its extremes fall well short of Lamb's (5 % of grains above 25° against "most"). Further limitations: the grain structure is locked to the registered hatch pattern of the deck, no surviving grains originate mid-build by nucleation at any density tested (on a 2° segmentation the coherent model's intragranular boundaries supply a 55 : 86 originating-to-established ratio against Lamb's $> 500 : 100$), the seed scatter of the population score is about 0.1°, and a 2° re-segmentation changes the simulated grain population.

1 Summary

- Population SROD statistics in GP700 are those of a finite-variance rotation walk: mean $\propto \sqrt{h}$ within the resolution of the published curves, $D \approx 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$, Maxwell width. The same statistics arise from any process with independent increments per unit height, so they constrain the process, not its mechanism. Lamb's example grains (2024 Fig. 4) rotate coherently along a path.
- Model, both SPPARKS additive apps: `misorientation_diffusion` D_v (per-voxel walk, inherited per hop) plus `misorientation_grain_walk` D_g ρ `path` (coherent per-grain walk, one Gaussian rotation-vector increment per unit of inheritance path length, shared by every voxel of the grain, correlated between levels with persistence ρ). Orientation is inherited from one parent voxel and never averaged. The model is an at-capture proxy for what Lamb attribute to post-solidification plasticity.

- Deck v10: D_v 0.065 deg^2/µm, D_g 0.010 deg^2/µm, $\rho = 0.9$ (correlation length ≈ 30 µm of path), calibrated on the 2024 originating curve (in-sample RMS 0.34°). Held-out: established curve RMS / bias 0.28 / −0.11° and 0.47 / 0.25° in two builds (v8, per-voxel only: 2.99 / 2.98°); 2026 Fig. 7a mode 1.6 → 4.7° over 5–115 µm against v10's 0.8 → 4.8° and v8's 1.8 → 5.8°.
- Per grain (96 filtered grains): drift of the mean orientation median 7.2° (v8 4.2°), spread 3.4° (v8 6.1°). Against Lamb: accumulation rate mean 0.84 / max 1.92° per 30 µm (2026: 1.7 / 5.8); grains with maximum SROD > 25°: 5 % (2024: "most"); per-layer mean at 300 µm median 10.8°, max 18° (2024 example grains: 20–25°).
- The constants are per unit of growth path (measured path factor 1.47 at 3 µm, 1.68 at 1.5 µm) and first-order resolution independent: with ρ and D_g rescaled for the voxel size the deck reproduces Lamb at the experiment's 1.5 µm voxel with a bias of 0.64° (per-voxel term alone: 0.41°); the excess over the 3 µm result grows from 3 % at 45 µm to 23 % at 282 µm, and rescaling the constants by the measured path-factor ratio (1.47/1.68) reduces the bias to 0.18°.
- Limitations: hatch-locked grain structure (boundary autocorrelation 0.6–0.7 at the 80 µm lag; a random per-layer hatch offset leaves the SROD results unchanged and the period only partly reduced); no surviving mid-build origins at nucleation 6.5×10^{12}–8.5×10^{13} 1/m^3 (0–7 grains per run against Lamb's > 500); a 2° re-segmentation gives 139 filtered grains of ESD 29 µm (spin labels: 96, 148 µm) with the SROD curve within 0.6°; seed scatter of the population score ≈0.05° RMS on the proxy and a second full build within 0.3°; the ⟨001⟩ fraction within 15° of BD is 0.15–0.19 against a random 0.10 and has no measured target.

2 Experimental methods (Lamb et al. 2024, 2026)

Material and process. CoNi-base superalloy GammaPrint-700 (SB-CoNi-10), laser powder bed fusion on an SLM Solutions SLM125 at 176 W and 1137 mm/s, 80 µm spot (the papers give the spot size only; the deck reads it as $D4\sigma$), 80 µm hatch, 30 µm layers, bidirectional scan with 90° interlayer rotation (the 2024 Fig. 2c sketch shows the direction repeating every two layers and the scan directions at 45° to the block edges), single 10 mm raster, 200 °C plate preheat. The hatch registration between layers of the same direction is not stated. The 2026 paper supplies the thermophysical set used for its 3DThesis simulations: $k = 22.5$ W/(m K) (extrapolated to 950 °C), $\rho = 8650$ kg/m^3, $c_p = 600$ J/(kg K) (IN718 at 1000 °C), liquidus 1373 °C (1646 K), and an absorptivity of 0.9 selected to reproduce measured pool dimensions; the solidus, 1319 °C (1592 K), is from the 2024 paper.

3D datasets. 2024: TriBeam tomography (femtosecond-laser serial sectioning with EBSD on each slice), 1.5 µm voxels, 385 slices, a 1400 × 412 × 477 µm volume with over 9000 grains, reconstructed in DREAM.3D and segmented at a 2° misorientation threshold before any SROD calculation. 2026: a 120 × 120 × 169 µm volume at 0.1 × 0.1 × 1 µm resolving cells.

SROD. For each grain the reference orientation sits near the origination point, "the lowest location of a grain's boundary along the build direction", offset one voxel layer (1.5 µm) upward so the reference is not a boundary pixel; offsets up to 8 µm do not change the result. *Originating* grains have their lowest point inside the volume; *established* grains are cut by the dataset floor and referenced at the centroid of their footprint on the lowest slice. Grains shorter than 60 µm (two layers) or under 40 µm equivalent spherical diameter (ESD) are excluded, leaving over 500 originating and 100 established grains. Each grain is sliced along the build direction; each slice gives a SROD histogram fitted to a normal distribution, and the mean and the 90 % interval of that fit are plotted against height above the origin (Fig. 5a) or above the dataset floor (Fig. 5c).

Fig. 4 follows four grains (G1–G4) with IPF and SROD maps, the mean SROD per layer ("as large as 20–25°" after > 300 µm), and every voxel on the build-direction inverse pole figure with an arrow for the rotation path from bottom to top; G4 splits into two orientation clusters. The maximum SROD in any grain is 58.7°; "most columnar grains" contain over 25°. The 2026 paper's Fig. 7a is a heatmap of SROD against distance from the grain start over 0–120 µm for the grains of that volume (the paper does not say whether grains cut by the volume floor are included), with quoted average and maximum accumulation rates of 1.7 and 5.8° per 30 µm for bulk grains (2.5 and 7.2° for cells; the paper does not state how the average was formed); it also reports that cells switch ⟨100⟩ growth variants at fusion lines rather than rotating continuously. Both papers attribute the misorientation to plastic deformation: solidification shrinkage in the mushy zone and thermomechanical cycling in the heat-affected zone of subsequent layers (2024), and, in the 2026 paper, torsion and deflection of cells by their neighbours during growth in the pool, with shear strains of 0.2–3 % supported by synchrotron measurements; the last is an at-solidification mechanism.

3 Analysis of the experimental data

Fig. 5a and 5c of the 2024 paper were digitized from 300 dpi renders (`docs/srod_random_walk.py`; curves in `docs/lamb2023_fig5{a,c}_srod.json`); no digitization uncertainty is claimed beyond the line width, about 0.3°. The curves were compared with a rotation walk: independent small rotations of angle θ about random axes give

$$\mathrm{RMS}^2 = n\,\theta^2 = D\,d, \tag{1}$$

a Maxwell distribution (χ with three degrees of freedom) with mean $= 0.921\,\mathrm{RMS}$, and a 90 % interval of 1.38× the mean at every distance, which is also the value of 3.29σ/mean for the normal fits Lamb plot. D was fitted by least squares of $\mathrm{RMS}^2 = Dh$ to the digitized mean over 0–265 µm with the 0.921 factor applied.

Findings (Fig. 1): $D = 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$ fits the originating mean, and the measured width ratio is 1.3–1.5 at every height for the originating grains. The established grains are described by the same D only after adding an initial spread of 6.2° RMS in quadrature (one fitted parameter, the orientation scatter already present at the dataset floor); with it, their width ratio falls to 1.1–1.25 beyond 150 µm, as expected for a non-central distribution, and their mean runs above the model past 300 µm. Over 30–260 µm the square-root curve and Lamb's own reading (a fast first 30 µm, then 0.85° per layer) lie within the line width of each other; the two differ only below 30 µm, where the digitization is least reliable. The population statistics therefore establish independent finite-variance increments per unit height and their magnitude, not the mechanism, and not whether the increments belong to voxel lineages or to whole grains: each grain's coherent walk is one realisation, and the Maxwell width appears across grains either way. Fig. 4 supplies the missing information for four grains: the mean orientation moves along a path, so at least part of the accumulation is coherent across a grain.

4 Simulation methods

Code. SPPARKS, branch SURGE-kokkos, app `additive/texture/kk` (Kokkos/HIP on an AMD gfx1151 APU), one MPI rank. The CPU app `additive/texture` carries the same model and is the correctness reference; the temperature field is deterministic and bit-identical between the two (`examples/am_validation/`). Microstructure is not deterministic on the GPU, so every build is one realisation of its seed.

Thermal source. Moser Green's-function solution with the GP700 constants (`setup_temperature_source moser 176 0.9 22.5 4.335e-6 473 600 20e-6 20e-6`

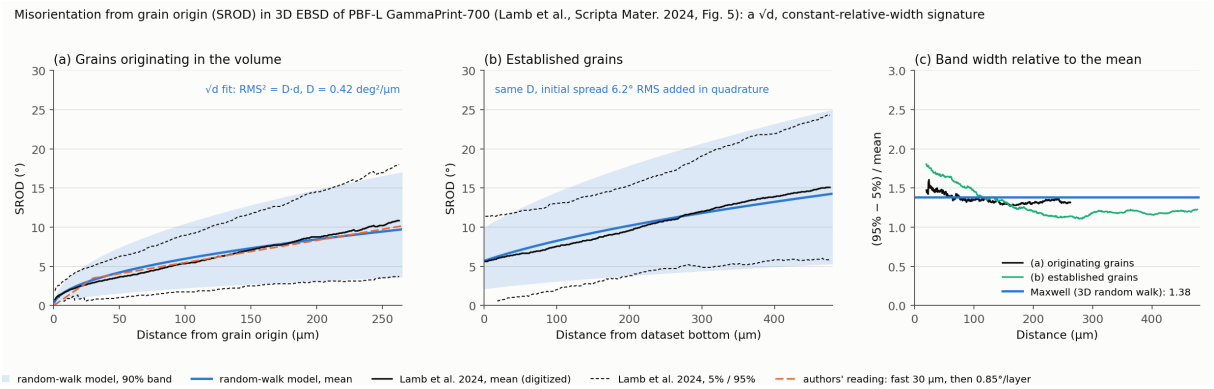


Figure 1: Lamb 2024 Fig. 5 digitized. (a) Mean SROD of originating grains with $\mathrm{RMS}^2 = Dh$, $D = 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$, and the authors' two-regime reading. (b) Established grains with the same D plus a fitted 6.2° RMS initial spread. (c) Width ratio (90 % interval over mean) against height for both classes; the Maxwell value is 1.38.

`10e-6`: power, absorptivity, conductivity, diffusivity, preheat, heat capacity, and the Gaussian source widths $\sigma_x, \sigma_y, \sigma_z$ of 20, 20, 10 µm), evaluated on the device for every site each step. Scan lines with `hatch 80e-6 N serpentine pause 8.8e-3`, the pause being the 10 mm track length over the scan speed; the fast-forward predictor skips idle gaps. Single-hatch check: pool 147 × 69 µm, thermal gradient 1×10^7–2.9×10^7 K/m at the fusion boundary (paper: 3×10^7 K/m).

Build. Box-per-layer stitch continuation (`run_layers.sh`): each layer is a separate SPPARKS run of a 180 µm window below the laser plane, started from the preheat and from the previous layer's microstructure. Domain 801 µm square (267 voxels at 3 µm), 10 hatches at $40 + 80k$ µm with lines 100 µm past both edges, 12 layers of 30 µm, 90° rotation per layer; layers of the same direction have their hatches at the same positions (registered), which is the deck's assumption where the papers are silent. Substrate: 150 µm of 50 µm Voronoi grains with uniform random orientations (`make_substrate.py`); five overlapping hatches remelt 80–100 µm, so the analysable solid begins at 105 µm (the analysis floor is set at 100 µm). Timestep 4 µs, twice the front-advance limit $\Delta x / V(T_l - T_s)$; 2 and 4 µs agreed on grain count and size within the run-to-run scatter. Calibration used a 600 µm, 7-hatch, 8-layer proxy at 3 min of loop time per box.

Solidification kinetics. Front velocity is a cubic polynomial in undercooling (the 304L stainless-steel fit shipped with the additive app) scaled ×3.5 (front at 25–35 K undercooling, about 0.56 m/s, keeping up with the 1.137 m/s scan); nucleation density $1.5 \times 10^{13}/\mathrm{m}^3$, critical undercooling $t_c = 15$ K with a 2 K standard deviation across sites. Parent selection per grain: candidate neighbours are grouped by grain, each grain weighted by the mobility $1 + c_2 \cos(c_3 \theta)$ of its mean orientation (the normalised sum of the candidate voxels' quaternions after flipping each to the same hemisphere) against the local thermal gradient (θ the angle between the nearest $\langle 100 \rangle$ of that mean orientation and the gradient; $c_2 = 2.8$, $c_3 = 2.5$, negative clamped to zero), then the parent voxel is drawn uniformly inside the chosen grain. Weighting among the voxels of one grain instead rotated grains toward $\langle 001 \rangle$ with height and is not used.

Misorientation model. Two terms, both applied at capture in the sample frame (Fig. 2; Fig. 3 contrasts them on one lineage tree and Fig. 4 shows their combination). The model is a phenomenological proxy: it places at solidification the rotation that Lamb attribute to shrinkage in the mushy zone and to cycling in the heat-affected zone of later layers. It therefore cannot represent any dependence of a voxel's SROD on the number of layers deposited above it, and a 12-layer build is compared here with material from the interior of a 10 mm block.

- *Per-voxel walk*, `misorientation_diffusion` D_v `30`: the captured voxel takes its parent's orientation rotated by $\theta = \sqrt{D_v \,\mathrm{hop}}$ about a random axis (hop = lattice distance to the parent, so D_v is per unit of growth path). A rotation past 30° would start a new grain (never reached). Independent lineages of one grain diverge, so this term supplies intragranular spread and no net drift.
- *Coherent grain walk*, `misorientation_grain_walk` D_g ρ `path`: every voxel carries its inheritance path length s in lattice units (per-site field d10: zero at melt and at nuclei, parent plus hop at capture, copied with the orientation on a smoothing flip, exchanged with ghosts and continued through the stitch). Between path levels k and $k+1$ every voxel of grain g receives the same rotation, a Gaussian rotation vector v_k of RMS $\sqrt{D_g \Delta x}$ whose sequence in k is Ornstein–Uhlenbeck, $v_k = \rho\, v_{k-1} + \sqrt{1-\rho^2}\, v'_k$, with v'_k hashed from (g, k) so the sequence is a pure function of grain and level, path independent and without stored state. The correlation length is $\approx \Delta x/(1-\rho)$ of path and the long-range accumulation is $D_g(1+\rho)/(1-\rho)$ per unit path; ρ is defined per lattice level, so a change of Δx requires $\rho' = 1-(1-\rho)\Delta x'/\Delta x$ to keep the correlation length and a corresponding rescaling of D_g (Section 6.8). This term supplies the coherent drift. A device cache of the recurrence makes it cost nothing measurable.

Boundary smoothing. One greedy steepest-descent pass per captured voxel (`smooth_greedy_multiproposal on`, nrefine 1) the step after capture, and `smooth_orientation copy`: a voxel that changes grain takes one random neighbour's orientation and path length from that grain. The default `average` (sign-aligned mean of the new grain's neighbours) cancels the fresh rotations of a voxel-scale mixed front, retains only 0.3–0.45 of the input D and widens the population band to 1.8; it is not used.

5 Simulation analysis methods

`gp700_analysis.py` on the spparks-micro-analysis tools: read the last stitch frame; solid means `i2 == 3` (solidified; unremelted substrate, `i2 == 1`, is excluded); crop to the central 400 µm laterally (300 µm on the proxy), grains cut by the crop faces kept with their cropped volume; label grains as connected components of the spin field, or optionally re-segment at a disorientation threshold (`-segment-deg`); grain volume, equivalent spherical diameter (ESD, from the cropped volume) and extents; Lamb's filters (height ≥ 60 µm, ESD ≥ 40 µm); origin = the grain's lowest solid voxel, whose quaternion is the reference (exact in the simulation, so no offset); grains crossing the analysis floor are established and referenced at their floor-slice centroid. SROD is the symmetry-reduced disorientation of every voxel from its grain's reference, binned by height above the origin or floor in 10 µm steps with mean, p_5, p_{95} and RMS per bin. Score: RMS of (simulated mean − Lamb Fig. 5a mean) over 25–240 µm, its bias, the band ratio $(p_{95}-p_5)$/mean averaged over the same bins, the fraction of solid within 15° of $\langle 001 \rangle$ parallel to the build direction (random: 0.102 by Monte Carlo), and grain counts. ESD values quoted for a build are volume-weighted medians over all grains of the crop (the ESD of the grain holding the median voxel); the number-weighted median ESD of the filtered grains is 75 µm in v10 and 72 µm in v8, and the filtered grains hold 96 % of the crop volume. Calibration: Sobol designs (scipy `qmc`; 16 + 8 points over D_v 0.2–2.0, nucleation 1×10^{13}–1×10^{14}, t_c 3–25 K, c_2 1–4, then 8 points over c_2 2–8 and nucleation 5×10^{12}–5×10^{13}) on the proxy against the Fig. 5a score fixed D_v for the per-voxel model; for the two-term model ρ was set from a target correlation length of 30 µm of path (the height of Lamb's initial fast regime), D_g from the requirement that the coherent term supply most of the long-range accumulation with the per-voxel term retained for intragranular spread ($D_g(1+\rho)/(1-\rho) : D_v \approx 3$), and the sum rescaled once to the Fig. 5a level; the Fig. 5c established curve and the 2026 Fig. 7a heatmap were not consulted.

`grain_paths.py` reproduces Lamb's Fig. 4 per grain and extends it to every filtered grain:

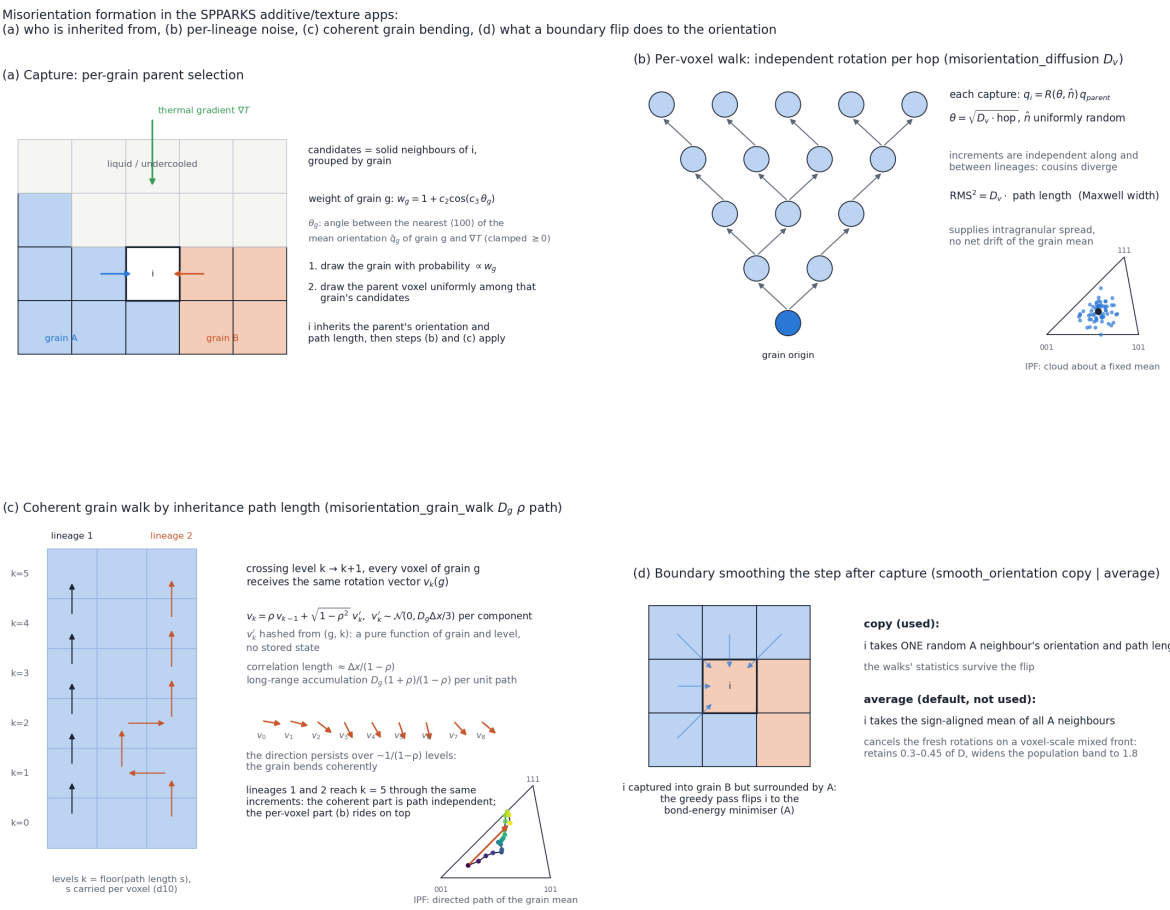


Figure 2: How orientation is assigned in the additive apps. (a) At capture the solid neighbours of the liquid voxel are grouped by grain, a grain is drawn with probability proportional to its mobility against the local thermal gradient, and the parent voxel is drawn uniformly within it. (b) The per-voxel walk rotates the inherited orientation by $\sqrt{D_v\,\text{hop}}$ about a random axis at every hop, so lineages of one grain diverge and the grain's mean does not move. (c) The coherent walk gives every voxel of a grain that crosses path level k the same rotation vector v_k, an Ornstein–Uhlenbeck sequence in k hashed from the grain and level, so the grain bends as a whole and the result is independent of the lineage path. (d) A boundary-smoothing flip copies one neighbour's orientation and path length (used) rather than the neighbours' average (default), which would cancel the fresh rotations.

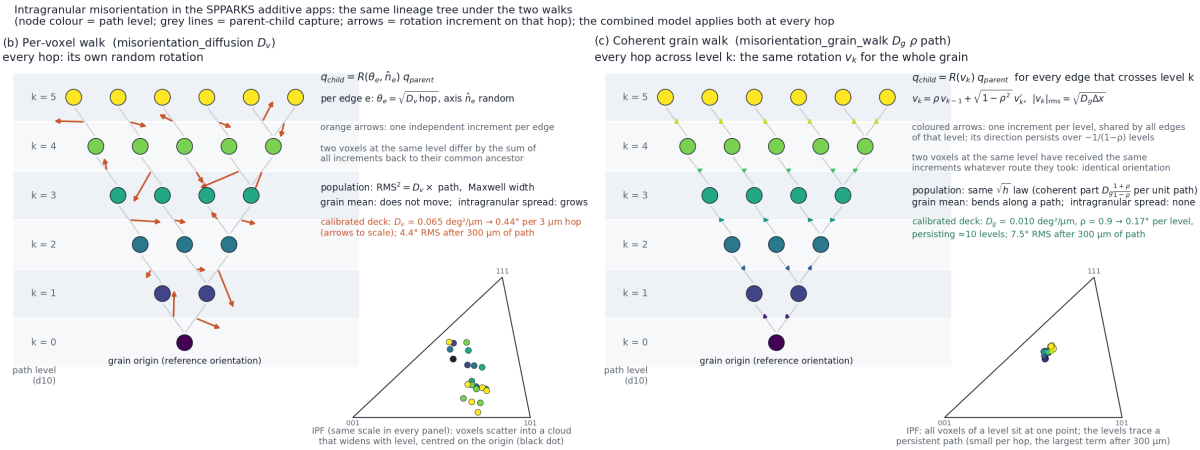


Figure 3: The two walks side by side on one lineage tree (grey lines: parent to child at capture; node colour: path level k; arrows: the rotation increment applied on that hop). (b) Per-voxel walk: each hop draws its own increment, so two voxels of the same level differ by every increment back to their common ancestor; in the inverse pole figure the grain becomes a cloud that widens with level while its mean stays at the origin. (c) Coherent grain walk: every hop that crosses level k applies the same vector v_k, whatever route the voxel took, so all voxels of a level share one orientation; v_k is correlated with v_{k-1} (persistence ρ), and in the inverse pole figure the levels trace a directed path away from the origin with no intragranular spread. The calibrated deck applies both at every hop. Arrows and inverse-pole-figure sketches are at the calibrated deck's per-hop magnitudes, $D_v = 0.065$ and $D_g = 0.010$ deg^2/µm, $\rho = 0.9$, $\Delta x = 3$ µm: 0.44° per hop for the per-voxel term against 0.17° per level for the coherent term, which is the smaller per hop but, persisting over about ten levels, the larger at long range (7.5 against 4.4° RMS after 300 µm of path).

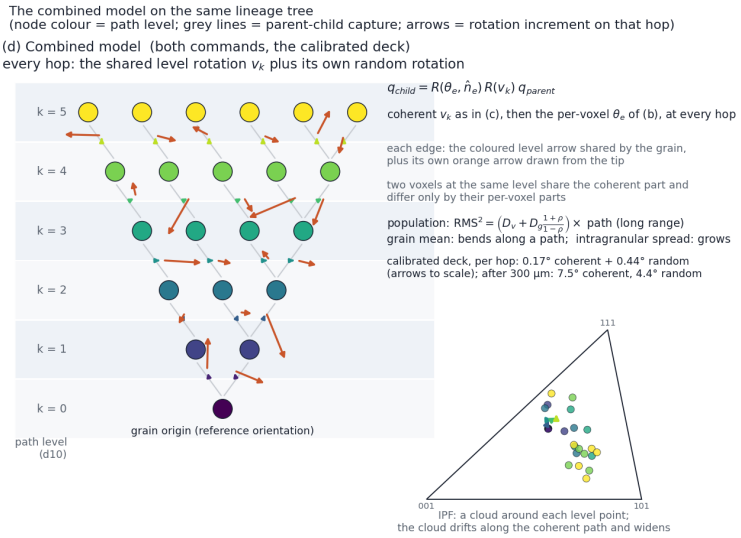


Figure 4: The combined model on the same lineage tree: every hop applies the level rotation v_k shared by the grain (coloured arrow) and then its own random rotation (orange arrow drawn from the coloured tip). Two voxels of one level share the coherent part and differ only by their per-voxel parts. In the inverse pole figure each level is a cloud around the corresponding point of the coherent path, so the grain mean drifts along the path while the intragranular spread widens. The long-range population law is RMS^2 $= (D_v + D_g(1+\rho)/(1-\rho)) \times$ path. Increments are drawn at the calibrated deck's per-hop magnitudes (0.17° coherent, 0.44° random); after 300 µm of path the coherent part reaches 7.5° RMS and the random part 4.4°.

IPF-Z and SROD maps on the xz slice through the grain, mean SROD per 10 µm slab, the per-slab mean orientation as a path on the build-direction inverse pole figure, the drift of the mean orientation from the bottom three slabs to the top three, the RMS spread of the top slabs about their mean, the maximum SROD, the per-layer mean at 300 µm, the accumulation rate (drift per 30 µm of height), the path length per unit height from d10, and the fraction of slabs in which the mean SROD rises. Both scripts use the same labelling, so the filtered-grain counts agree except that `grain_paths.py` drops grains with fewer than 20 voxels in their bottom or top slabs (97 vs 96 in v10, 88 vs 86 in v8). `hatch_periodicity.py` measures the boundary-voxel density across the crop, its autocorrelation at the hatch lag, and in-plane grain widths by line intercept; `origin_census.py` counts filtered grains that originate more than 60 µm above the first remelt floor.

6 Results

6.1 Calibration path

Table 1: Full builds against Lamb 2024 Fig. 5a (all-originating SROD, central 400 µm crop, one realisation each).

build	model	RMS / bias vs Fig. 5a	band ratio	per-grain drift / spread (median)
v2	per-voxel walk, D_v 0.42, voxel-level parent selection	2.2° / 3.6° (linear growth)	1.4–1.8	–
v5	per-grain parent selection, average smoothing, D_v 1.15	0.59° / −0.16°	1.79	–
v8	copy smoothing, D_v 0.22	0.49° / 0.17°	1.41	4.2 / 6.1°
v10	**+ coherent path walk D_g 0.010, ρ 0.9; D_v 0.065**	**0.34° / −0.28°**	**1.50**	**7.2 / 3.4°**

The Fig. 5a score is the calibration target, so the v8 and v10 rows are in-sample fits (v10 is set in bold as the working deck, not as the better-supported model), and their difference (0.49 vs 0.34°) is within the digitization line width; the population curve does not choose between the two models. v2's linear growth was selection drift among the voxels of one grain. v5 to v7 retained only 0.3–0.45 of the input D against height with a 1.8 band, both from the averaging smoothing rule; copy-mode smoothing (v8) put the population curve, shape and band on Lamb. v8's grains, however, spread without drifting, whereas Lamb's grains move along a path; the coherent grain walk (v10) supplies the drift. A memoryless coherent walk makes individual SROD(h) curves wander; correlating the rotation vector between levels (ρ) makes them rise along a directed path, and with Gaussian increments the population band stays at Lamb's value in both the ballistic and the diffusive regime.

6.2 Held-out comparisons

Table 2: Targets not used in calibration. Established class: simulated grains crossing a 240 µm floor, referenced at their floor-slice centroid, against Lamb Fig. 5c over 25–233 µm. 2026 Fig. 7a: voxel-weighted SROD distribution against distance from the grain start, digitized from the heatmap (the mode is the usable statistic; the density-weighted mean is raised by grains starting at the volume floor).

	v10	v8	Lamb
established vs Fig. 5c: RMS / bias	0.28° / −0.11°	2.99° / 2.98°	–
established band ratio	1.39	1.49	1.4 (2024, above 150 µm: 1.1–1.25)
established mean at 25 / 100 / 160 µm	5.5 / 7.3 / 8.8°	9.2 / 10.5 / 11.7°	6.2 / 7.5 / 8.9°
2026 Fig. 7a mode at 5 / 30 / 60 / 115 µm	0.8 / 1.8 / 2.8 / 4.8°	1.8 / 2.8 / 3.8 / 5.8°	1.6 / 2.9 / 3.7 / 4.7°
2026 Fig. 7a mean at 5 / 60 / 115 µm	1.3 / 3.6 / 5.8°	2.1 / 4.4 / 6.4°	4.9 / 5.2 / 6.2°

The established class is the discriminating test. The per-voxel model's floor-referenced grains start 3° too high because their lineages have already diverged across the grain by the time they cross the floor, whereas the coherent model, whose intragranular spread is small, reproduces the

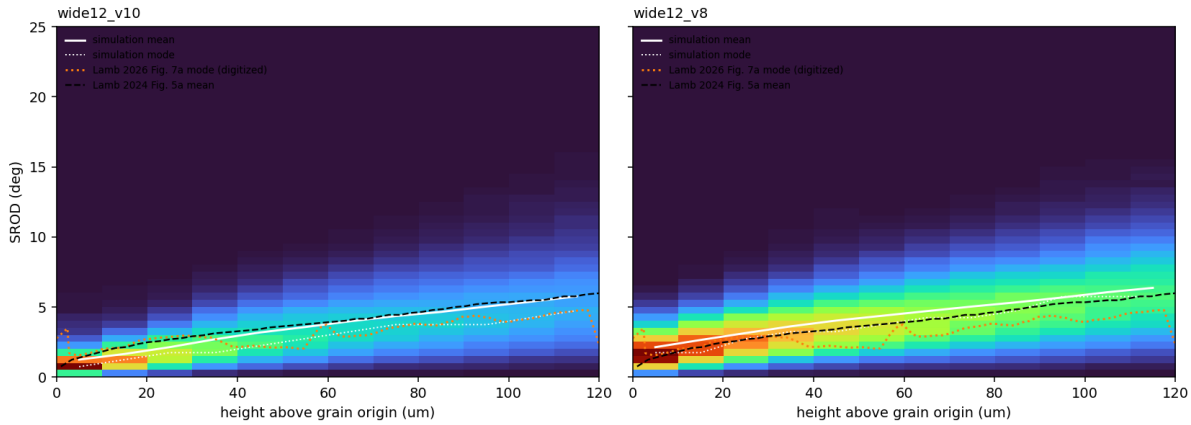


Figure 5: Voxel-weighted SROD against height above the grain origin over the first 120 µm for the filtered originating grains of v10 (left) and v8 (right), with the digitized mode of Lamb 2026 Fig. 7a and the 2024 Fig. 5a mean.

curve to 0.28° in the first build and 0.47° (bias 0.25°) in the second, with the Maxwell band. The result depends on where the floor is placed, because the model's intragranular spread grows with path from the origin: floors at 200, 240 and 300 µm give biases of −0.65, −0.11 and 0.44°, so the agreement is at the 0.5° level rather than the 0.1° the best floor suggests. The 2026 heatmap is a weaker test. Its density-weighted mean (about 5° from the grain start) carries no stated digitization uncertainty and is flat because the volume contains grains that start at or below its floor, as the established class does in 2024; only the mode is used here. Against the mode, v10 runs 30–50 % low over the first 60 µm (0.8 vs 1.6° at 5 µm) and meets it by 115 µm, while v8 tracks it within 0.2° to 60 µm and overshoots by 115 µm; the RMS of the modal residuals is 0.8° for v10 and 0.6° for v8. The 2026 mode at 5 µm (1.6°) is the sub-30 µm information the 2024 curves lack; it sits above the pure $D = 0.42$ walk (1.2°) and above v10, i.e. the experiment accumulates faster than either over the first two voxel layers of the coarse dataset.

6.3 v10 against Lamb Fig. 5a

The fitted D against height exceeds the deck constants times the measured path factor by 15 % (v10) and 28 % (v8): the walk's own excess over the height index (diagonal hops, and for the coherent term the ballistic regime below the correlation length, where accumulation runs faster than the long-range rate) adds to the geometric path factor.

Table 3: v10: D_v 0.065 deg^2/µm, D_g 0.010 deg^2/µm, ρ 0.9 path, copy smoothing, nrefine 1, Voronoi substrate, nucleation 1.5×10^{13}/m^3, t_c 15 K, c_2 2.8, Δt 4 µs, 3 µm; single realisation.

	v10	v8 (per-voxel only)	Lamb 2024
SROD RMS deviation, 25–240 µm (in-sample)	0.34°	0.49°	–
mean bias	−0.28°	0.17°	–
mean SROD at 54 / 204 / 402 µm	3.6 / 8.2 / 12.1°	4.4 / 8.5 / 12.3°	3.8 / 8.8 / – (data end at 262 µm)
band ratio, 25–240 µm average	1.50	1.41	1.4
D-fit vs height	0.43 deg^2/µm	0.41 deg^2/µm	0.42
(constants × path factor: $(D_v + D_g(1+\rho)/(1-\rho)) \times 1.47$)	0.37	0.32	–
grains passing filters (400 µm crop)	97	86	> 600 (full volume)
ESD p_{10} / p_{50} / p_{90}	76 / 148 / 170 µm	73 / 151 / 182 µm	–
⟨001⟩ within 15° of BD (random 0.10)	0.15	0.19	not reported

6.4 Per-grain orientation paths

The coherent term changes the per-grain picture in the way Lamb's Fig. 4 requires: the mean orientation of a v10 grain drifts more than it spreads, the drift scales with height (about 2° at

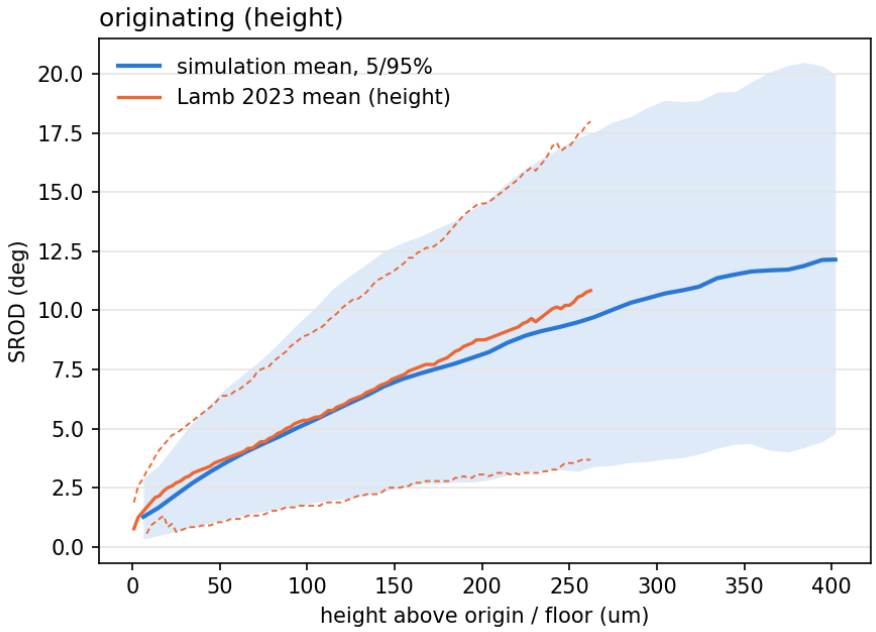


Figure 6: v10, all-originating SROD against height above the origin (central 400 µm crop, floor 100 µm): simulated mean with the 5/95 band (blue) against Lamb 2024 mean and 90 % interval (orange).

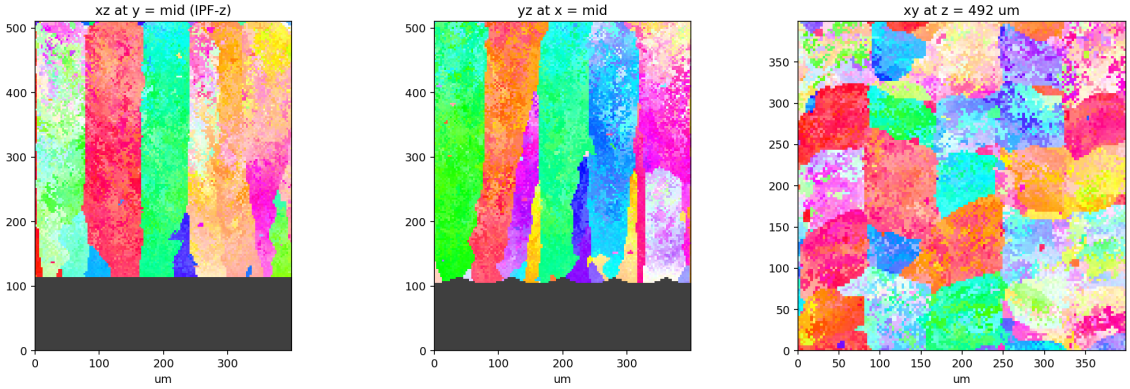


Figure 7: v10 IPF-Z colouring: xz and yz sections through the crop centre, and the xy section at 492 µm. Dark grey is the unremelted substrate. The boundaries at 80 µm intervals coincide with the hatch centrelines (Section 6.5).

Table 4: Every filtered grain of the two full builds (`grain_paths.py`), medians with p_{10} / p_{90} for the drift, against the per-grain quantities Lamb report.

	v10 (96 grains)	v8 (86 grains)	Lamb
drift of the mean orientation, bottom → top	7.2° (1.7 / 13.5)	4.2° (2.0 / 7.5)	directed paths (4 grains)
spread about the top-slab mean (RMS)	3.4°	6.1°	–
drift ÷ spread	1.66	0.60	–
accumulation rate, drift per 30 µm: mean / max	0.84 / 1.92°	0.57 / 1.47°	1.7 / 5.8° (2026, bulk grains)
per-layer mean SROD at 300 µm: median / p_{90} / max	10.8 / 15.6 / 18.0°	9.9 / 12.3 / 14.4°	“as large as” 20–25° (G1–G4)
maximum SROD in the grain: median / p_{90} / max	15.0 / 23.2 / 33.4°	16.0 / 26.7 / 34.6°	up to 58.7°
grains with maximum SROD > 25°	5 %	17 %	“most”
fraction of slabs with rising mean SROD	0.79	0.78	–
path length per unit height (d10)	1.47	–	–

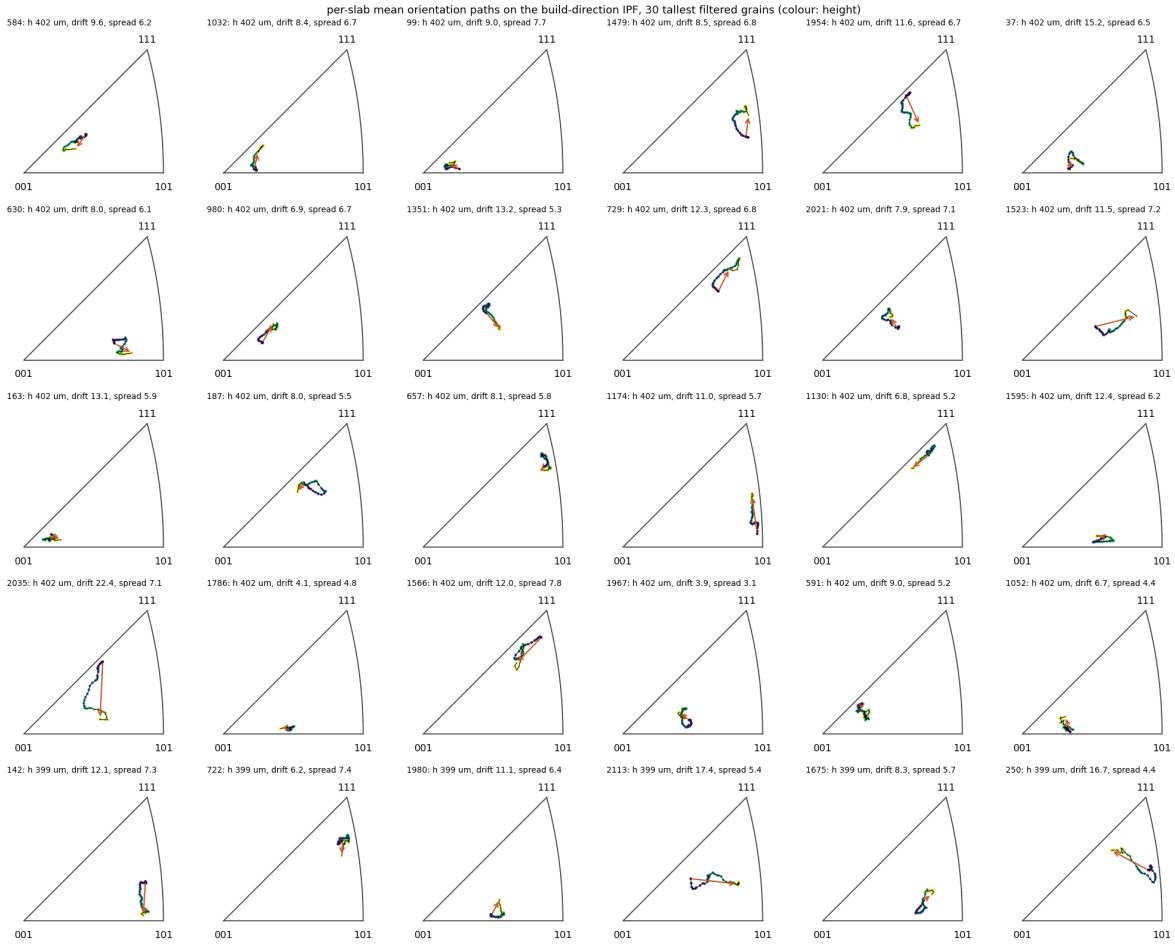


Figure 8: v10: per-slab mean orientation of the 30 tallest filtered grains as paths on the build-direction inverse pole figure (colour: height; orange arrow: bottom to top). All 30 span the full analysable height and are substrate continuations.

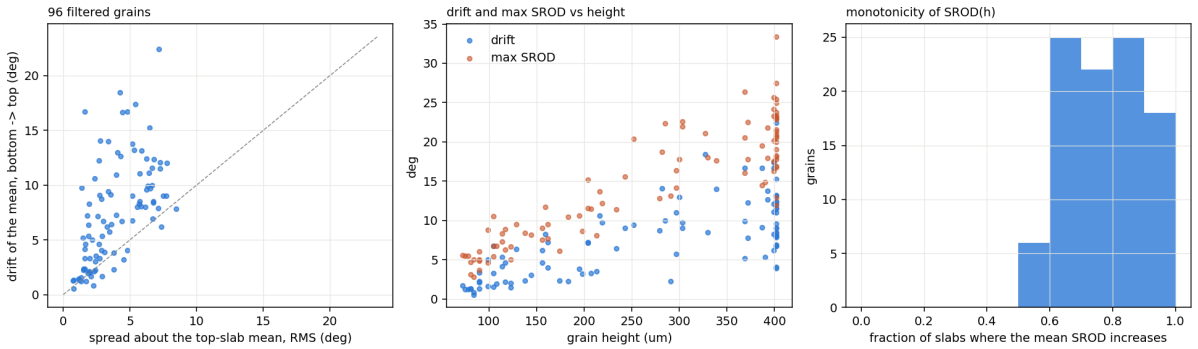


Figure 9: v10, all 96 filtered grains: drift against spread (dashed: equal), drift and maximum SROD against grain height, and the monotonicity of the per-grain SROD(h).

100 µm, 8–15° at 400 µm), and the paths are directed; v8's grains spread more than they drift at every height. Against Lamb's numbers, however, both models fall short. The accumulation rate of the v10 grains (drift of the mean orientation per 30 µm of height, averaged over grains) is half the 2026 average and its maximum is a third of theirs, if the paper's average is a comparable per-grain quantity; the per-layer means at 300 µm reach 18° at most against "as large as" 20–25° for Lamb's four example grains, which were chosen as illustrations and are not a population statistic; and 5 % of the v10 grains exceed 25° anywhere, against "most" in the experiment (a maximum-SROD statistic that in EBSD is sensitive to boundary and misindexed voxels, so the experimental side is an upper estimate). Lamb's own figures are not mutually consistent (a mean above 25° over 300 µm in the text against 11° at 260 µm in Fig. 5a); the population curves are taken as the target and the per-grain numbers as indicative. The population mean is matched, so the experimental population must carry a much heavier tail, a minority of grains rotating several times faster than the median, than a single Gaussian D_g produces; the 10–15° secondary band of the 2026 heatmap and the split grain G4 point the same way. The paths have no preferred direction on the triangle, as the isotropic rotation vector implies; the experimental direction is not reported. Monotonicity does not separate the models (0.79 vs 0.78): about a fifth of the grains reverse over 100 µm scales in both. The per-grain drift-to-spread ratio confirms the construction ($\sqrt{D_g(1+\rho)/(1-\rho)/D_v} = 1.7$) rather than an experimental value; Lamb's within-slice widths, which would fix it, are not published.

6.5 Grain structure and the hatch pattern

The grain boundaries of both full builds fall at 80 µm intervals in x and in y: the boundary-density autocorrelation at the 80 µm lag is 0.61 / 0.70 (x / y) in v10 and 0.58 / 0.71 in v8, and 43–62 % of the spectral power of the boundary density sits at the hatch period (`hatch_periodicity.py`, Fig. 10). In-plane grain widths by line intercept have medians of 30, 39 and 51 µm at 200, 300 and 405 µm height (p_{90} 72–84 µm). The cause is the deck's registered hatch pattern: layers n and $n+2$ have identical scan lines, so the pool centrelines, where the last liquid solidifies, recur at the same positions and select the same columns, and the grains become $\sim 80 \times 80$ µm columns spanning the build, which also compresses the ESD distribution (p_{10}–p_{90} 76–170 µm). Lamb describe serrated, irregular boundaries. The papers do not state the hatch registration of the SLM125 build. As a sensitivity test the driver gained a per-layer hatch offset (`HATCH_OFFSET=random`: a seeded uniform shift in $[-40, 40)$ µm along the hatch normal, so layers n and $n+2$ no longer register), and the v10 settings were run on the proxy with it against a registered proxy of the same seed. The offset lowers the boundary autocorrelation at the hatch lag from 0.46 / 0.45 to 0.34 / 0.48 (x / y) and the spectral power at the period from 0.47 / 0.57 to 0.52 / 0.39; grain widths (median 27–36 vs 30–33 µm), ESD (p_{50} 120 vs 123 µm), the SROD score (0.30° / −0.14°, band 1.50) and the per-grain drift and spread (5.3 / 3.6° vs 6.0 / 3.8°) are unchanged within the seed scatter, while the ⟨001⟩ fraction rises from 0.21–0.25 to 0.30 (one realisation). A normalised autocorrelation of the height-integrated boundary density is insensitive to phase, so it cannot separate registration from the pool spacing itself, which imprints an 80 µm period on every layer; the unchanged grain widths and ESD under the offset are the evidence that the column selection is not registration-driven, and the SROD results do not depend on it. Whether the experimental grains carry the same imprint cannot be read from the published sections.

6.6 Grain definition

Re-segmenting the simulated volume at Lamb's 2° disorientation threshold (`-segment-deg 2`) changes the grain population. v10: 411 000 labels, 139 pass Lamb's filters (spin labels: 96), ESD p_{50} 29 µm, two grains span 300 µm; the Fig. 5a score stays within 0.6° (bias −0.45°, band 1.49), and the per-grain drift / spread become 5.5 / 1.9°. v8: 758 000 labels, 54 filtered grains

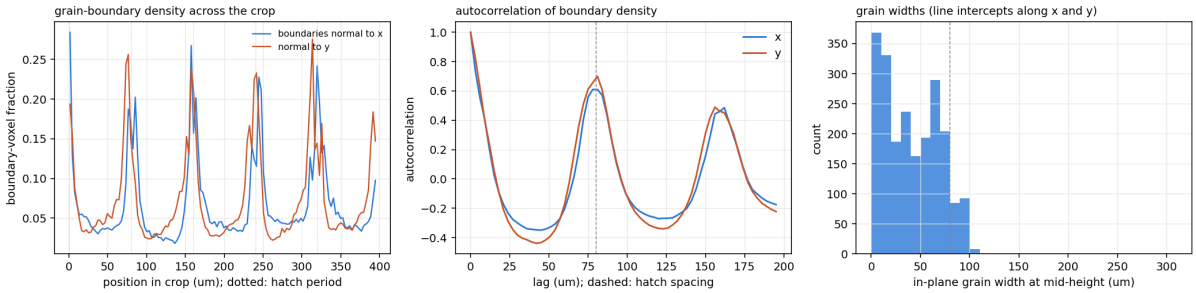


Figure 10: v10: grain-boundary density across the crop with the hatch period marked, its autocorrelation, and the in-plane grain-width distribution at mid-height.

of ESD p_{50} 8 µm, no grain taller than 165 µm, score 1.19°. The per-voxel walk alone does not survive a 2° segmentation as columnar grains; the coherent model keeps its SROD curve but not its grain size, because the simulated same-grain neighbour disorientation (median 2.4° in v10) sits at the threshold. The simulated field carries voxel-scale orientation noise that the EBSD does not (the 2.4° neighbour disorientation is a property of independent-lineage inheritance at this D_v and would be lower only with a smaller per-voxel term), so the two segmentations are not equivalent; the spin labelling is the one used elsewhere in this report. A threshold sweep locates the equivalence (Table 5): at 3° the coherent model recovers its columnar population (97 filtered grains, 37 taller than 300 µm, ESD p_{50} 128 µm) with a Fig. 5a score of 0.25°, whereas the per-voxel model needs 5°; since the EBSD orientation noise is a few tenths of a degree against a median same-grain neighbour disorientation of 2.4° (v10) and 3.8° (v8) in the simulation, a 3° threshold on v10 plays the role of Lamb's 2°. The equivalence moves with the voxel size: the per-voxel step scales as $\sqrt{\Delta x}$, so at 1.5 µm the same-grain neighbour disorientation falls to 1.9° (2.1° at 3 µm on the same proxy) and a 2° segmentation of the 1.5 µm v10 run keeps its columnar grains (59 filtered grains, ESD p_{50} 79 µm, tallest grains spanning the build, Fig. 5a RMS 0.16°) where the 3 µm proxy at 2° fragments (ESD p_{50} 45 µm, RMS 0.69°); at 3° the two resolutions agree (ESD p_{50} 102 vs 101 µm). No orientation smoothing is applied before segmentation; the labels come from the raw voxel quaternions, and the only smoothing in the pipeline is the simulation's boundary-smoothing pass on grain membership during the run. The 2° segmentation does, however, change the grain-class picture of the next section: on it, 55 of the 141 filtered v10 grains (13 % of the filtered volume) originate more than 60 µm above the remelt floor, because the coherent walk builds low-angle boundaries inside columnar grains that a 2° threshold reads as new grains, as Lamb's G4 does; v8 gives none.

Table 5: Re-segmentation of the two full builds at disorientation thresholds of 2, 3 and 5° (`spparks_micro_analysis.grain_segmentation`) against the spin labelling; central 400 µm crop, floor 100 µm. ESD is the volume-weighted median over all labels.

build	labelling	labels	filtered	taller than 300 µm	ESD p_{50}	RMS / bias vs Fig. 5a	band ratio
v10	spin	2504	97	48	148 µm	0.34° / −0.28°	1.50
v10	2°	411 240	139	2	29 µm	0.60° / −0.45°	1.49
v10	3°	248 414	97	37	128 µm	0.25° / −0.11°	1.61
v10	5°	104 195	91	39	144 µm	0.57° / 0.54°	1.78
v8	spin	2496	86	41	151 µm	0.49° / 0.17°	1.41
v8	2°	758 358	54	0	8 µm	1.19° / −0.98°	1.36
v8	3°	498 507	126	11	16 µm	1.51° / −0.89°	1.40
v8	5°	271 319	94	30	128 µm	0.93° / 0.71°	1.84

6.7 Grain origins

With the spin labelling, 3 of the 100 filtered v10 grains and 3 of the 94 filtered v8 grains originate more than 60 µm above the remelt floor (0.5 % of the filtered volume): the columnar grains are epitaxial continuations of substrate grains, so the simulated "originating" SROD is an established-grain SROD referenced at the remelt floor. A census over 26 existing proxy runs spanning nucleation densities of 6.5×10^{12}–8.5×10^{13} $1/\mathrm{m}^3$ and critical undercoolings of 3–23 K finds 0–7 filtered grains per run originating more than 60 µm above the floor (0–4 % of the filtered volume); Lamb's volume has over 500 such grains against 100 established. No region of the tested parameter space produces surviving mid-build origins by nucleation; nucleated grains are overgrown. Lamb's originating grains are products of a 2° segmentation, and on the same segmentation the coherent model does produce mid-build origins (previous section), so the like-for-like comparison is 55 : 86 (v10, 2°) against Lamb's $> 500 : 100$; the model is short by a factor of about three in the ratio, not absent. Analysing v8 on the full 801 µm domain, the central 400 µm, the central 200 µm, or without the never-remelted top layer changes the SROD bias by under 0.2° and the band by under 0.03.

6.8 Resolution

The per-voxel model (v8 deck) was run at Lamb's 1.5 µm voxel on the proxy with the same D_v and geometry: SROD RMS / bias 0.49° / 0.41° against 0.44° / 0.03° at 3 µm, band 1.43 vs 1.41, same grain size and texture, with the finer lattice accumulating 10–18 % more per unit height because it resolves a slightly longer growth path. For the coherent term, ρ is per lattice level: keeping the 30 µm correlation length at 1.5 µm takes $\rho' = 0.95$ and, to keep the long-range accumulation, $D'_g = D_g(1+\rho)(1-\rho')/((1-\rho)(1+\rho')) = 0.0049\,\mathrm{deg}^2/\mu\mathrm{m}$. With that rescaling the v10 deck at 1.5 µm on the proxy gives Fig. 5a RMS / bias 0.74° / 0.64° (3 µm, three seeds: 0.35 ± 0.05 / $+0.05 \pm 0.11$°), band 1.42 (1.53), per-grain drift / spread 7.8 / 4.4° (6.6 / 3.9°), maximum SROD median 15.6° (13.7), accumulation rate mean 1.00° per 30 µm (0.84), ESD p_{50} 115 µm (123), and a measured path length per unit height of 1.68 (1.47–1.49). The mean SROD runs above the 3 µm value by an amount that grows with height: 3 % at 45 µm (3.4 vs 3.3°), 10 % at 205 µm (9.6 vs 8.7) and 23 % at 282 µm (12.4 vs 10.1). To test whether the longer resolved growth path (1.68 vs 1.47) is the cause, the run was repeated with D_v and D_g additionally scaled by $1.47/1.68 = 0.875$: the bias falls from 0.64° to 0.18° (RMS 0.30°, band 1.49, drift / spread 7.1 / 4.4°, path factor 1.76), within two standard deviations of the 3 µm seed scatter, and the excess over the 3 µm curve shrinks to −4 % at 45 µm, +7 % at 205 µm and +12 % at 282 µm. The path factor therefore accounts for most of the resolution dependence, with a residual of order 10 % at the top of the build that grows with height and is not explained. The two-term model is resolution independent to first order once ρ, D_g and the per-path constants are rescaled with the voxel size and the measured path factor.

6.9 Seed scatter

The seed-to-seed scatter of the population score is about 0.05° RMS and 0.1° to 0.2° in bias, so the in-sample difference between the two models on the Fig. 5a curve (0.50 vs 0.35°) is at the two-sigma level and not decisive, as stated above; the band ratio, grain count and texture do not separate the models at all. The per-grain drift and spread separate them by more than five standard deviations in each direction. A second full 12-layer v10 build with a different seed gives, against the first: Fig. 5a RMS 0.21° / bias 0.00° (first: 0.34 / −0.28°), band 1.46 (1.50), 97 filtered grains (97), ESD p_{50} 156 µm (148), $\langle 001 \rangle$ 0.20 (0.15), established-class RMS / bias vs Fig. 5c 0.47° / 0.25° (0.28 / −0.11°), per-grain drift / spread medians 6.6 / 3.2° (7.2 / 3.4), per-layer mean at 300 µm median 9.9° (10.8), maximum SROD median 12.7° (15.0), path factor 1.49 (1.47). The second build also contains one grain that reaches 49° and accumulates 5.4° per

Table 6: Three seeds each of the v8 and v10 settings on the 600 µm proxy (8 layers, central 300 µm); mean ± standard deviation over seeds.

	v8 settings (per-voxel only)	v10 settings (coherent + per-voxel)
SROD RMS vs Fig. 5a	$0.50 \pm 0.05°$	$0.35 \pm 0.05°$
mean bias	$+0.17 \pm 0.19°$	$+0.05 \pm 0.11°$
band ratio	1.45 ± 0.05	1.53 ± 0.07
grains passing filters	52 ± 2	55 ± 3
⟨001⟩ within 15°	0.23 ± 0.02	0.23 ± 0.02
per-grain drift, median	$3.9 \pm 0.3°$	$6.6 \pm 0.5°$
per-grain spread, median	$6.7 \pm 0.7°$	$3.9 \pm 0.4°$
per-grain maximum SROD, median	$16.6 \pm 0.1°$	$13.7 \pm 0.3°$

30 µm, the only simulated grain so far in the range of Lamb's fastest; the per-grain extremes are seed-sensitive where the medians are not.

6.10 Texture

Neither paper quantifies a ⟨001⟩ fibre; the 2026 paper discusses both ⟨001⟩- and ⟨111⟩-oriented grains. A Sobol study over the mobility contrast c_2 and nucleation density with D fixed gave the ⟨001⟩-within-15° fraction flat at 0.22 below $c_2 \approx 5$ and rising by 0.0125 per unit c_2 above it, with the SROD score unaffected; on the full build $c_2 = 9$ gave 0.22 against 0.19 at $c_2 = 2.8$. The coherent walk lowers the fraction (v10: 0.15) because it carries grains out of the cone. Against a random baseline of 0.102, the builds sit at 1.5–2.2× random.

7 Findings and open issues

1. Population SROD in GP700 has the statistics of a finite-variance rotation walk, $D \approx 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$ against height with a Maxwell width; individual grains rotate along a path. The statistics do not identify the mechanism, and the published curves do not distinguish $\sqrt{h}$ from the authors' two-regime reading above 30 µm; the 2026 mode of 1.6° at 5 µm indicates faster accumulation below 30 µm than the single-D walk gives.
2. The two-term model reproduces the calibration curve and, without further tuning, the established-grain curve to within 0.5° (floor-placement dependent), which the per-voxel model alone fails by 3°; against the 2026 modal trajectory the two models are comparable. Per grain the model produces directed paths but accumulates at half Lamb's average rate with far smaller extremes: the experiment has a heavy tail of fast-rotating grains that a single Gaussian D_g does not give.
3. Deck constants are per unit of growth path; the measured path length is 1.47 times the height. Both terms are resolution independent to first order once ρ and D_g are rescaled with Δx, with the measured path factor (1.47 at 3 µm, 1.68 at 1.5 µm) carrying most of the residual resolution dependence.
4. The simulated grain structure is locked to the deck's registered hatch pattern; the experimental hatch registration is unknown.
5. Grain classes: with spin labels no surviving grains originate mid-build at any tested nucleation density; with Lamb's 2° segmentation the coherent model gives an originating-to-established ratio of 55 : 86, against Lamb's $> 500 : 100$, because its intragranular boundaries read as new grains.
6. A 2° re-segmentation of the simulation changes its grain population; the two grain definitions are not equivalent, and the per-voxel term sets the intragranular neighbour disorientation that decides it.

7. D_g : D_v and ρ are fixed here by construction and by the held-out curves only loosely; Lamb's within-slice widths, per-grain rate distribution and path directions would fix them.
8. Not yet tested: the undercooling dependence of D_v; a heavy-tailed or grain-dependent D_g.

8 Reproduction

```
. /home/tron/spack/share/spack/setup-env.sh && spack env activate spparks-dev
# binary: build-verify/spparks (commit 6a47de77 or later); template = the v10 deck
E=examples/gp700_moser; R=/home/tron/gp700_runs
$E/run_layers.sh build-verify/spparks $R/wide12_v10 12 12345 267 10 4e-6 $R/
    substrate_800um_50um_d10.st
/home/tron/miniconda3/envs/WORK/bin/python $E/gp700_analysis.py $R/wide12_v10/gp700.st
     \
    --out $R/wide12_v10/analysis_floor100 --floor-um 100 --crop 200 600 200 600
/home/tron/miniconda3/envs/WORK/bin/python $E/grain_paths.py $R/wide12_v10/gp700.st \
    --out $R/wide12_v10/grain_paths --crop 200 600 200 600 --floor-um 100 --grid 30
/home/tron/miniconda3/envs/WORK/bin/python $E/hatch_periodicity.py $R/wide12_v10/gp700
    .st --out $R/wide12_v10/hatch --crop 200 600 200 600
```

Reproducible artefacts: the digitized curves `docs/lamb2023_fig5{a,c}_srod.json` (the 2024 paper; file names carry the manuscript year) and `docs/lamb2026_fig7a_srod.json`, the scripts above and `docs/srod_random_walk.py`, the per-grain tables `docs/gp700_results_v{8,10}/grain_stats.csv` and the review-response outputs in `docs/gp700_results_review/`. Key commits: fa8930f3 (per-voxel walk), ad23c1a5 (per-grain selection), 1a9712b3 (kk kernel parity), da2fad29 (`smooth_orientation`), b6e4ba5f / 47546146 / 60b2d711 / 6a47de77 (`misorientation_grain_walk`), f5d79cd3 (v10 deck), 71824e43 / 21de639b (review analyses). The development log is `docs/GP700_MOSER_SROD_DECK_PLAN.md`; the first review and the response are `docs/SROD_REPORT_REVIEW.md` and `docs/SROD_REPORT_RESPONSE.md`. This file is the reference version; `docs/SROD_ANALYSIS_REPORT.md` and the published HTML page mirror it.

Sources. J. Lamb, K. M. Pusch, A. T. Polonsky, S. A. J. Forsik, N. Zhou, A. D. Dicus, R. Geurts, M. P. Echlin, T. M. Pollock, Scripta Materialia 239 (2024) 115770 (3D-EBSD and SROD of PBF-L GammaPrint-700; manuscript dated 2023); J. Lamb et al., Metallurgical and Materials Transactions A 57A (2026) 2430 (cellular structure, intragranular SROD, 3DThesis parameters).